RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

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Year / Branch / Section : 2026 /IT/ E

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College Roll No. : 251001027

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INDEX PAGE

Exp.No	Name of the Experiment	Date of Completion
1	Finding Complexity using Counter Method	
2	Finding Complexity using Counter method	
3	Finding Complexity using Counter Method	
4	Finding Complexity using Counter Method	
5	Finding Complexity using Counter Method	
6	1-G-Coin Problem	
7	2-G-Cookies Problem	
8	3-G-Burger Problem	
9	4-G-Array Sum max problem	
10	5-G-Product of Array elements-Minimum:	
11	1-Number of Zeros in a Given Array	
12	2-Majority Element:	
13	3-Finding Floor	
14	4-Two Elements sum to x	
15	5-Implementation of Quick Sort	
16	1-DP-Playing with Numbers	
17	2-DP-Playing with chessboard	
18	3-DP-Longest Common Subsequence	

19	4-DP-Longest non-decreasing Subsequence:	
20	1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity	
21	2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity	
22	3-Print Intersection of 2 sorted arrays -O(m*n)Time Complexity,O(1) Space Complexity	
23	4-Print Intersection of 2 sorted arrays -O(m+n)Time Complexity,O(1) Space Complexity:	
24	5-Pair with Difference-O(n^2)Time ComplexityO(1) Space Complexity:	
25	6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity:	

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Saturday, 1 August 2026, 11:01 AM
State	Finished
Completed on	Saturday, 1 August 2026, 11:17 AM
Time taken	16 mins 28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
void function (int n)
{
    int counter=0;
    int i= 1;
    int s =1;
    counter+=2;
    while(s <= n)
    {
        counter+=1;
         i++;
         counter+=1;
         s += i;
        counter+=1;
     }
     counter+=1;
     printf("%d",counter);
 }
int main(){
    int n;
    scanf("%d",&n);
    function(n);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Saturday, 1 August 2026, 11:18 AM
State	Finished
Completed on	Saturday, 1 August 2026, 11:29 AM
Time taken	11 mins 35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
void func(int n)
{
    int counter=0;
    if(n==1)
    {
        counter+=1;
      //printf("*");
     counter+=1;
    }
    else
    {
     for(int i=1; i<=n; i++)

     {

       for(int j=1; j<=n; j++)
       {
           counter+=1;

          //printf("*");
          counter+=1;
          //printf("*");
          counter+=1;
          counter+=1;
          break;
       }
       counter+=1;

     }
     counter+=1;
   }
   counter+=1;
   printf("%d",counter);
 }
```

```
int main(){
    int n;
    scanf("%d",&n);
    func(n);
    return 0;

}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Saturday, 1 August 2026, 11:30 AM
State	Finished
Completed on	Saturday, 1 August 2026, 11:45 AM
Time taken	14 mins 4 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>

void Factor(int num)
 {
     int c=0;
    for (int i = 1; i <= num;++i)
    {
    c+=1;
    c+=1;
     if (num % i== 0)
        {

          //printf("%d ", i);
          c+=1;
        }
     }
     c+=1;
     printf("%d",c);
 }
  int main(){
    int n;
    scanf("%d",&n);
    Factor(n);

}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Friday, 7 August 2026, 10:29 PM
State	Finished
Completed on	Friday, 7 August 2026, 10:39 PM
Time taken	10 mins 3 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
void function(int n)
{
    int c= 0;
    c++;
    for(int i=n/2; i<n; i++){
        c++;
        for(int j=1; j<n; j = 2 * j){
            c++;
            for(int k=1; k<n; k = k * 2){
                c++;
                c++;
            }
            c++;
        }
        c++;
    }
    c++;
    printf("%d",c);
}
int main(){
    int n;
    scanf("%d",&n);
    function(n);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Friday, 7 August 2026, 10:40 PM
State	Finished
Completed on	Friday, 7 August 2026, 10:48 PM
Time taken	8 mins 33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
void reverse(int n)
{
   int c=0;
   c++;
   int rev = 0, remainder;
   c++;
   while (n != 0)
    {
        c++;
        remainder = n % 10;
        c++;
        rev = rev * 10 + remainder;
        c++;
        n/= 10;
        c++;
    }

    //print(rev);
    c++;
    printf("%d",c);

}

int main(){
    int n;
    scanf("%d",&n);
    reverse(n);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:13 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:13 AM
Time taken	44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.

Input Format:

Take an integer from stdin.

Output Format:

print the integer which is change of the number.

Example Input :

64

Output:

4

Explanaton:

We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int v, count = 0;
    int coins[] = {1000, 500, 100, 50, 20, 10, 5, 2, 1};

    scanf("%d", &v);

    for (int i = 0; i < 9; i++) {
        count += v / coins[i];
        v = v % coins[i];
    }

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	49	5	5	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:15 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:15 AM
Time taken	24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten i burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3 burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$.

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

Input Format

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

Output Format

Print: Minimum number of kilometers needed to run to burn out the calories

Sample Input

3

5 10 7

Sample Output

76

For example:

Test	Input	Result
Test Case 1	3 1 3 2	18

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<math.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] < arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    int s=0;
    for(int i=0;i<n;i++){
        s+=(pow(n,i))*arr[i];
    }
    printf("%d",s);
}
```

	Test	Input	Expected	Got	
✔	Test Case 1	3 1 3 2	18	18	✔
✔	Test Case 2	4 7 4 9 6	389	389	✔
✔	Test Case 3	3 5 10 7	76	76	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

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Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:16 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:16 AM
Time taken	30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    return (*(int *)a - *(int *)b);
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    qsort(arr, n, sizeof(int), compare);

    long long sum = 0;

    for (int i = 0; i < n; i++)
        sum += (long long)arr[i] * i;

    printf("%lld", sum);

    return 0;
}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔

	Input	Expected	Got	
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:16 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:17 AM
Time taken	27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int asc(const void *a, const void *b) {
    return *(int *)a - *(int *)b;
}

int desc(const void *a, const void *b) {
    return *(int *)b - *(int *)a;
}

int main() {
    int n;
    scanf("%d", &n);

    int a[n], b[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (int i = 0; i < n; i++)
        scanf("%d", &b[i]);

    qsort(a, n, sizeof(int), asc);
    qsort(b, n, sizeof(int), desc);

    long long sum = 0;

    for (int i = 0; i < n; i++)
        sum += (long long)a[i] * b[i];

    printf("%lld", sum);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔

	Input	Expected	Got	
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:18 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:20 AM
Time taken	1 min 12 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the index of the first occurrence of 0
int findFirstZero(int arr[], int low, int high) {
    if (high >= low) {
        int mid = low + (high - low) / 2;

        // Check if mid is the first 0
        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
            return mid;
        }

        // If mid is 1, the first 0 is in the right half
        if (arr[mid] == 1) {
            return findFirstZero(arr, mid + 1, high);
        }

        // If mid is 0 but not the first one, search the left half
        return findFirstZero(arr, low, mid - 1);
    }

    // Return -1 if no 0 is found in the array
    return -1;
}

int main() {
    int m;

    // Read the size of the array
    if (scanf("%d", &m) != 1) {
        return 0;
    }

    // Dynamically allocate memory for the array
    int *arr = (int *)malloc(m * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read m elements line by line
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    // Find the first zero index
    int firstZeroIndex = findFirstZero(arr, 0, m - 1);

    // If index is -1, there are no zeroes
    if (firstZeroIndex == -1) {
        printf("0\n");
    } else {
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:20 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:20 AM
Time taken	37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the majority element
int majorityElement(int* nums, int numsSize) {
    int candidate = 0;
    int count = 0;

    for (int i = 0; i < numsSize; i++) {
        // If count is 0, pick the current element as the candidate
        if (count == 0) {
            candidate = nums[i];
        }

        // Change count based on whether current element matches candidate
        if (nums[i] == candidate) {
            count++;
        } else {
            count--;
        }
    }

    return candidate;
}

int main() {
    int n;

    // Read the size of the array
    while (scanf("%d", &n) == 1) {
        int* nums = (int*)malloc(n * sizeof(int));
        if (nums == NULL) {
            return 1;
        }
```

```
35
36          // Read the array elements
37          for (int i = 0; i < n; i++) {
38              scanf("%d", &nums[i]);
39          }
40
41          // Find and print the majority element
42          int result = majorityElement(nums, n);
43          printf("%d\n", result);
44
45          // Free memory for the current testcase
46          free(nums);
47      }
48
49      return 0;
50  }
51
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:21 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:21 AM
Time taken	37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the floor of x using Divide and Conquer
int findFloor(int arr[], int low, int high, int x) {
    int floorVal = -1; // Stores the best candidate found so far

    while (low <= high) {
        int mid = low + (high - low) / 2;

        // If x is exactly present, it is its own floor
        if (arr[mid] == x) {
            return arr[mid];
        }

        // If mid element is smaller than x, it's a potential floor
        if (arr[mid] < x) {
            floorVal = arr[mid];  // Record this as a candidate
            low = mid + 1;        // Look for a larger value in the right half
        }
        // If mid element is greater than x, look in the left half
        else {
            high = mid - 1;
        }
    }

    return floorVal;
}

int main() {
    int n, x;

    // Read the size of the array
    if (scanf("%d", &n) != 1) return 0;

    // Dynamically allocate memory for the array
    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) return 1;

    // Read n elements line by line
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Read the value of x
    scanf("%d", &x);

    // Find and print the floor value
    int result = findFloor(arr, 0, n - 1, x);
    printf("%d\n", result);

    // Clean up allocated memory
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:22 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:22 AM
Time taken	44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Recursive Divide and Conquer function to find the pair
int findPairSum(int arr[], int left, int right, int x, int *ele1, int *ele2) {
    // Base case: if pointers cross, no such pair exists
    if (left >= right) {
        return 0;
    }

    int currentSum = arr[left] + arr[right];

    // If target sum is found, record the elements and return success
    if (currentSum == x) {
        *ele1 = arr[left];
        *ele2 = arr[right];
        return 1;
    }

    // Divide step: reduce search space depending on current sum
    if (currentSum < x) {
        return findPairSum(arr, left + 1, right, x, ele1, ele2); // Search right half
    } else {
        return findPairSum(arr, left, right - 1, x, ele1, ele2); // Search left half
    }
}

int main() {
    int n, x;

    // Read the size of the array
    if (scanf("%d", &n) != 1) return 0;

    // Dynamically allocate memory for the array
    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) return 1;

    // Read n elements line by line
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Read the target sum value x
    scanf("%d", &x);

    int ele1 = 0, ele2 = 0;

    // Find pair using the divide and conquer helper
    if (findPairSum(arr, 0, n - 1, x, &ele1, &ele2)) {
        printf("%d\n", ele1);
        printf("%d\n", ele2);
    } else {
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Monday, 31 August 2026, 10:23 AM
State	Finished
Completed on	Monday, 31 August 2026, 10:24 AM
Time taken	1 min 27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>
#include <stdlib.h>

// Function to swap two elements
void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

/*
   This function takes the last element as pivot, places
   the pivot element at its correct position in sorted array,
   and places all smaller elements to left of pivot and all
   greater elements to right of pivot.
*/
int partition(int arr[], int low, int high) {
    int pivot = arr[high]; // Choosing the last element as pivot
    int i = (low - 1);     // Index of smaller element

    for (int j = low; j < high; j++) {
        // If current element is smaller than or equal to pivot
        if (arr[j] <= pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    // Place pivot in its correct position
    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}

// The main function that implements QuickSort
void quickSort(int arr[], int low, int high) {
    if (low < high) {
        // pi is partitioning index
        int pi = partition(arr, low, high);

        // Separately sort elements before and after partition
        quickSort(arr, low, pi - 1);
        quickSort(arr, pi + 1, high);
    }
}

int main() {
    int n;

    // Read the number of elements
    if (scanf("%d", &n) != 1) {
        return 0;
    }

```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:17 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:26 PM
Time taken	9 mins 14 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    long long dp[n + 1];

    dp[0] = 1;

    for (int i = 1; i <= n; i++) {
        dp[i] = dp[i - 1];

        if (i >= 3) {
            dp[i] += dp[i - 3];
        }
    }

    printf("%lld", dp[n]);

    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔

	Input	Expected	Got	
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:21 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:22 PM
Time taken	50 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3

1 2 4

2 3 4

8 7 1

Output:

19

Explanation:

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

Input Format

First Line contains the integer n

The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n][n];
    int dp[n][n];

    // Read the chessboard
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    // Starting cell
    dp[0][0] = a[0][0];

    // First row
    for (int j = 1; j < n; j++) {
        dp[0][j] = dp[0][j - 1] + a[0][j];
    }

    // First column
    for (int i = 1; i < n; i++) {
        dp[i][0] = dp[i - 1][0] + a[i][0];
    }

    // Fill remaining cells
    for (int i = 1; i < n; i++) {
        for (int j = 1; j < n; j++) {
            if (dp[i - 1][j] > dp[i][j - 1])
                dp[i][j] = dp[i - 1][j] + a[i][j];
            else
                dp[i][j] = dp[i][j - 1] + a[i][j];
```

```
        }
    }

    // Maximum path value
    printf("%d", dp[n - 1][n - 1]);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 4 2 3 4 8 7 1	19	19	✔
✔	3 1 3 1 1 5 1 4 2 1	12	12	✔
✔	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:22 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:23 PM
Time taken	48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int main() {
    char s1[100], s2[100];

    scanf("%s", s1);
    scanf("%s", s2);

    int m = strlen(s1);
    int n = strlen(s2);

    int dp[m + 1][n + 1];

    // Initialize first row and first column
    for (int i = 0; i <= m; i++) {
        dp[i][0] = 0;
    }

    for (int j = 0; j <= n; j++) {
        dp[0][j] = 0;
    }

    // Build the DP table
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {

            if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            }
            else {
                if (dp[i - 1][j] > dp[i][j - 1])
                    dp[i][j] = dp[i - 1][j];
                else
                    dp[i][j] = dp[i][j - 1];
            }
        }
    }

    printf("%d", dp[m][n]);

    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:23 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:24 PM
Time taken	48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    int dp[n];

    // Read the sequence
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        dp[i] = 1;
    }

    // Dynamic Programming
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++) {

            if (a[j] <= a[i] && dp[j] + 1 > dp[i]) {
                dp[i] = dp[j] + 1;
            }
        }
    }

    // Find maximum value in dp
    int max = dp[0];

    for (int i = 1; i < n; i++) {
        if (dp[i] > max) {
            max = dp[i];
        }
    }

    printf("%d", max);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:27 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:28 PM
Time taken	1 min 20 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int slow = a[0];
    int fast = a[0];

    // Find intersection point
    do {
        slow = a[slow];
        fast = a[a[fast]];
    } while (slow != fast);

    // Find duplicate number
    slow = a[0];

    while (slow != fast) {
        slow = a[slow];
        fast = a[fast];
    }

    printf("%d", slow);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:28 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:29 PM
Time taken	40 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    int count[n + 1];

    // Initialize count array
    for (int i = 0; i <= n; i++) {
        count[i] = 0;
    }

    // Read elements
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        count[a[i]]++;
    }

    // Find duplicate
    for (int i = 1; i <= n; i++) {
        if (count[i] > 1) {
            printf("%d", i);
            break;
        }
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:29 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:30 PM
Time taken	48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;

        scanf("%d", &n1);

        int a[n1];

        for (int i = 0; i < n1; i++) {
            scanf("%d", &a[i]);
        }

        scanf("%d", &n2);

        int b[n2];

        for (int i = 0; i < n2; i++) {
```

```
        for (int i = 0; i < n2; i++) {
            scanf("%d", &b[i]);
        }

        int i = 0, j = 0;
        int first = 1;

        while (i < n1 && j < n2) {

            if (a[i] == b[j]) {
                if (!first)
                    printf(" ");

                printf("%d", a[i]);
                first = 0;

                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:30 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:31 PM
Time taken	51 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;

        scanf("%d", &n1);

        int a[n1];
        for (int i = 0; i < n1; i++) {
            scanf("%d", &a[i]);
        }

        scanf("%d", &n2);

        int b[n2];
        for (int i = 0; i < n2; i++) {
            scanf("%d", &b[i]);
        }
```

```
        int i = 0, j = 0;

        while (i < n1 && j < n2) {

            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:31 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:36 PM
Time taken	4 mins 38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0;
    int j = 1;

    while (i < n && j < n) {

        if (i == j) {
            j++;
        }
        else if (a[j] - a[i] == k) {
            printf("1");
            return 0;
        }
        else if (a[j] - a[i] < k) {
            j++;
        }
        else {
            i++;
        }
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Barath S 251001027-2025-IT

B2

Started on	Wednesday, 7 October 2026, 10:36 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:38 PM
Time taken	1 min 38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0;
    int j = 1;

    while (i < n && j < n) {

        if (i == j) {
            j++;
        }
        else if (a[j] - a[i] == k) {
            printf("1");
            return 0;
        }
        else if (a[j] - a[i] < k) {
            j++;
        }
        else {
            i++;
        }
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course